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SALIENCY DETECTION ON LIGHT FIELD DISPLAYS

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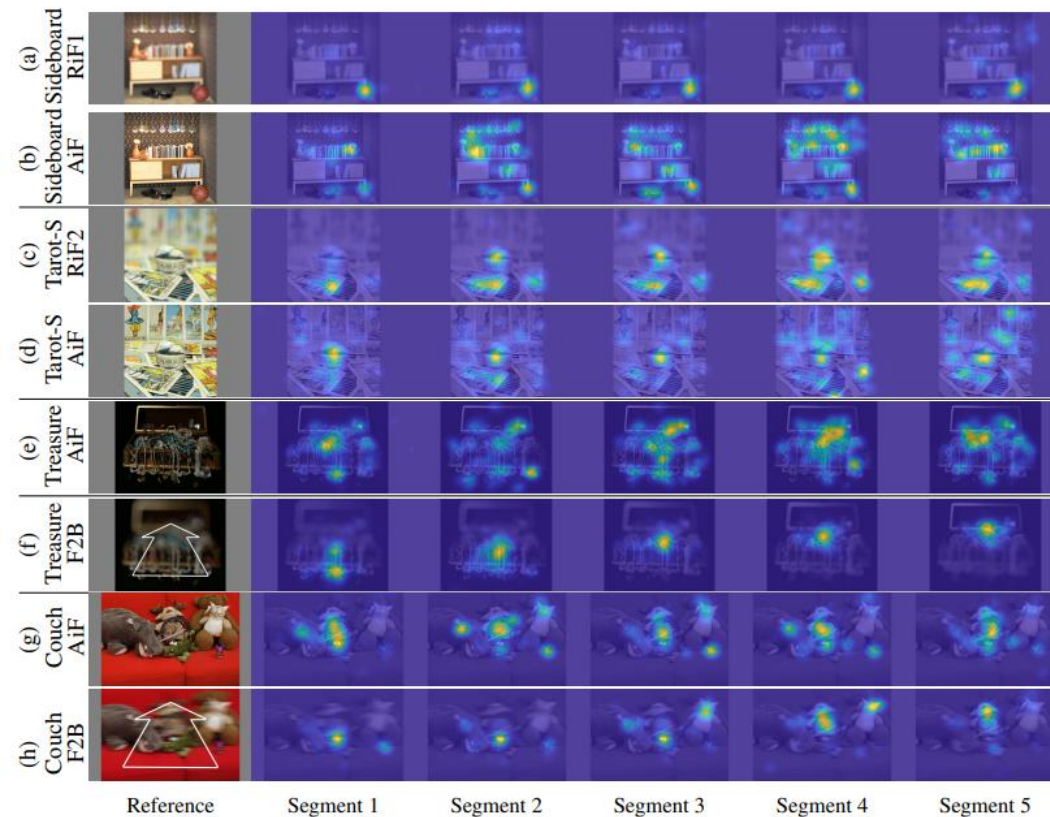
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Saliency detection

Saliency detection is the task **of identifying the most attention-grabbing regions or objects in a scene**, enabling **scene understanding**.



Light Field Imaging

Light field (LF) images capture both the **spatial** and **angular** information of light rays, allowing **continuous parallax** rendering and realistic **depth perception**.



LF devices can **reconstruct realistic and complex 3D scenes**.

Problem Statement

- Projection-based LF displays suffer from **aliasing artifacts** due to limited **angular resolution**.
- **Blurring filters** have been used to reduce these artifacts.
- **User studies** revealed that viewers often **preferred unfiltered aliasing** over blurred images.



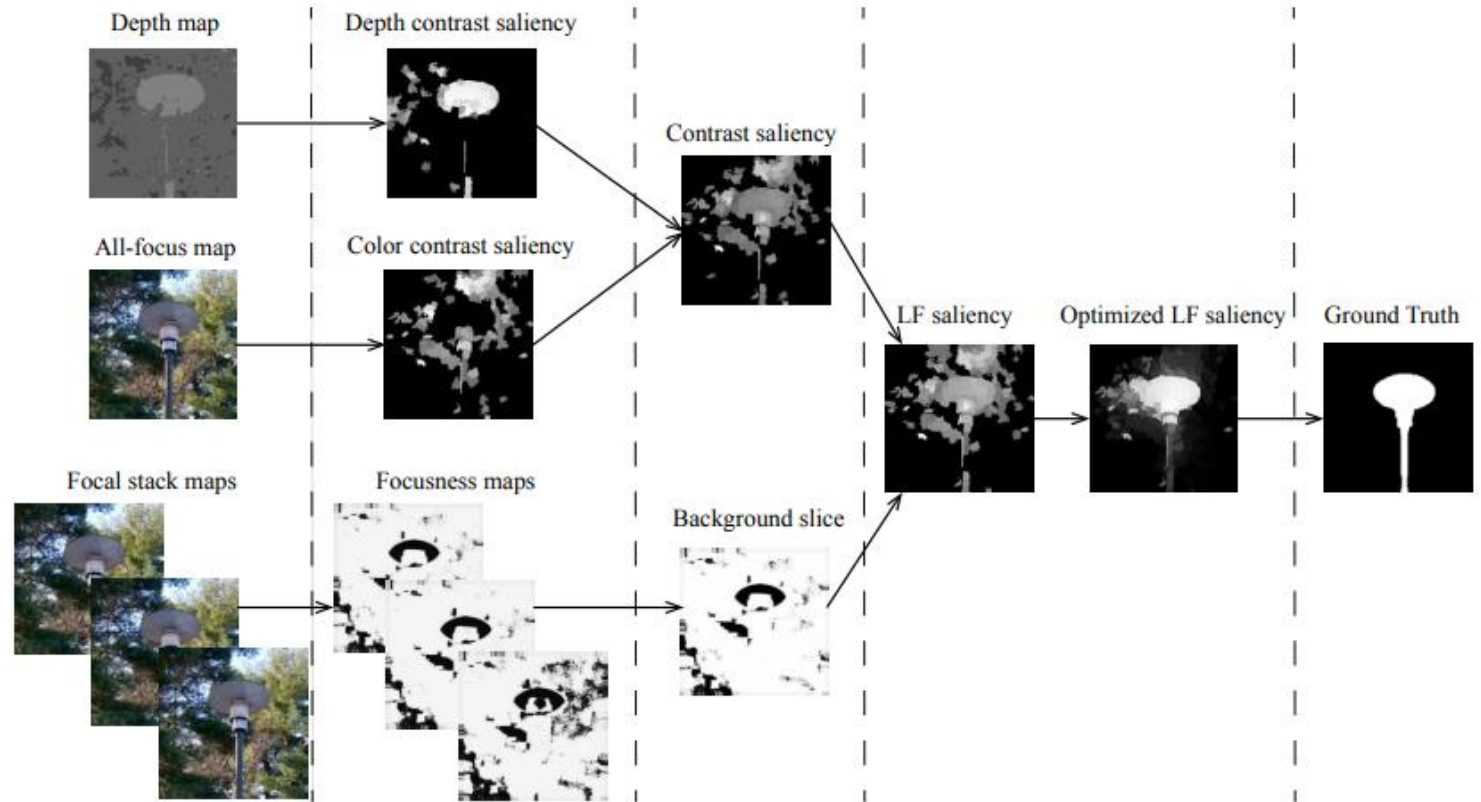
An example of projection-based LF display, Holografika Hologvizio 722RC

Contributions

- **Implemented and compared four saliency methods** (DILF, WSC, Multi-Cue, LF-Tracy) on a **custom LF dataset**.
- **Illustrated the superiority of LF-Tracy (data-driven approach)** in producing **sharper** and **defined** saliency maps for LF images.
- Highlighted **limitations and future directions** for modeling human-like attention in LF images.

Signal processing approaches

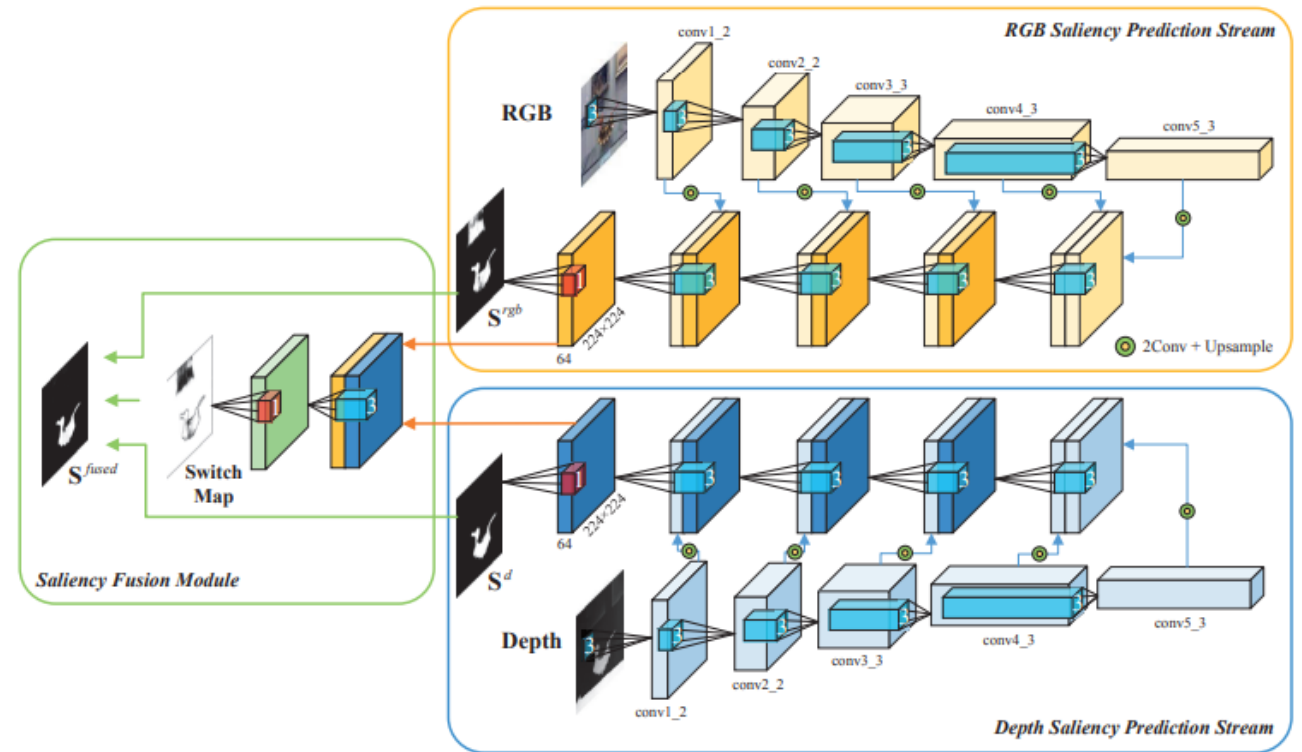
- Focusness
- Depth estimation
- Objectness
- Simple Linear Iterative Clustering (SLIC) algorithm



Limitations: sensitive to lighting, cluttered backgrounds and occlusion.

Data-driven approaches

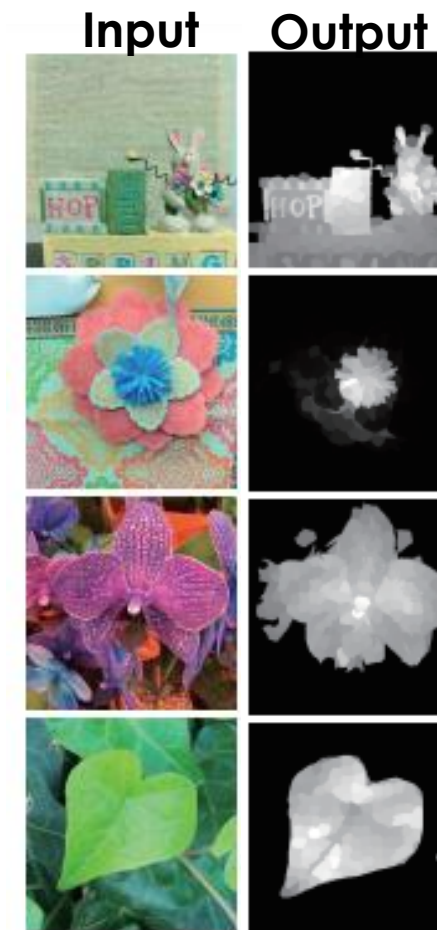
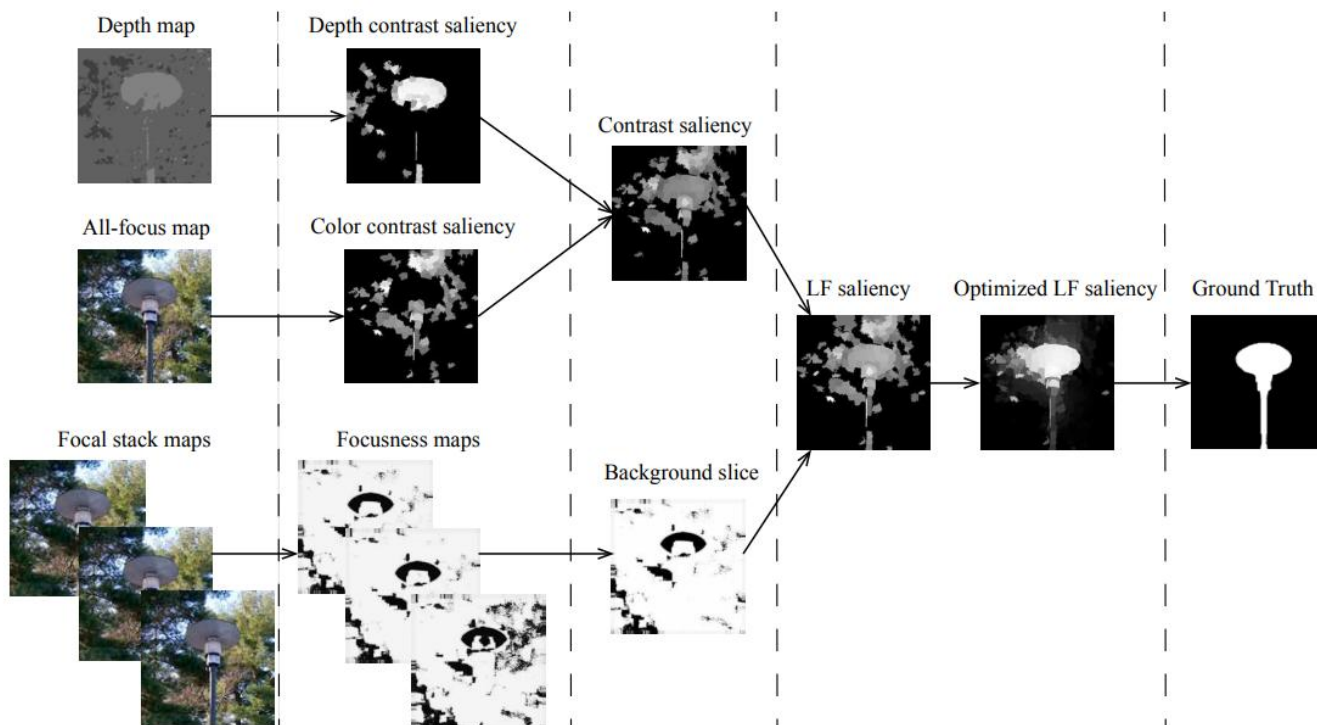
- Multistream neural architectures.
- Process RGB and depth features in parallel.
- RCNN, attention mechanisms, adversarial training.



Limitations: limited annotated LF datasets for supervised training.

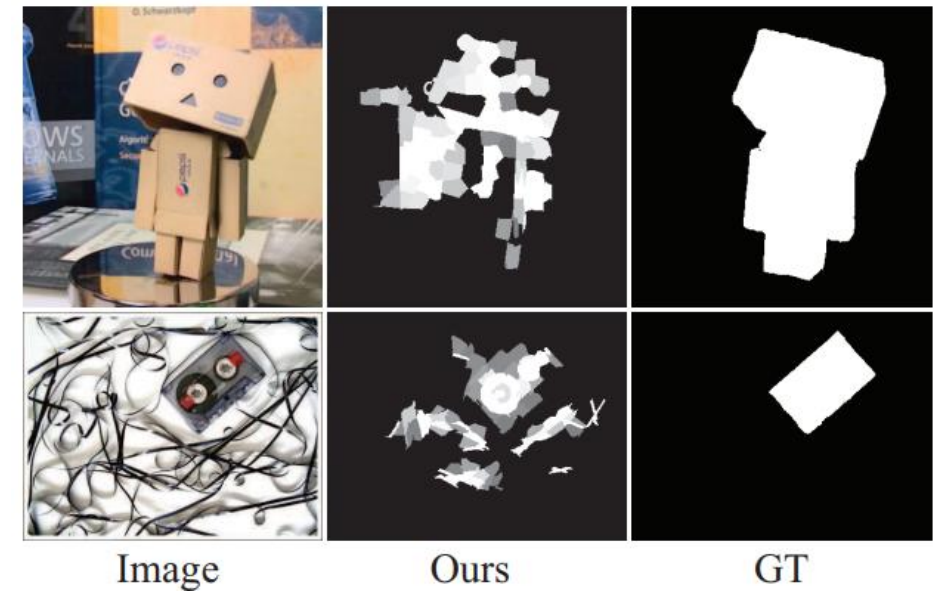
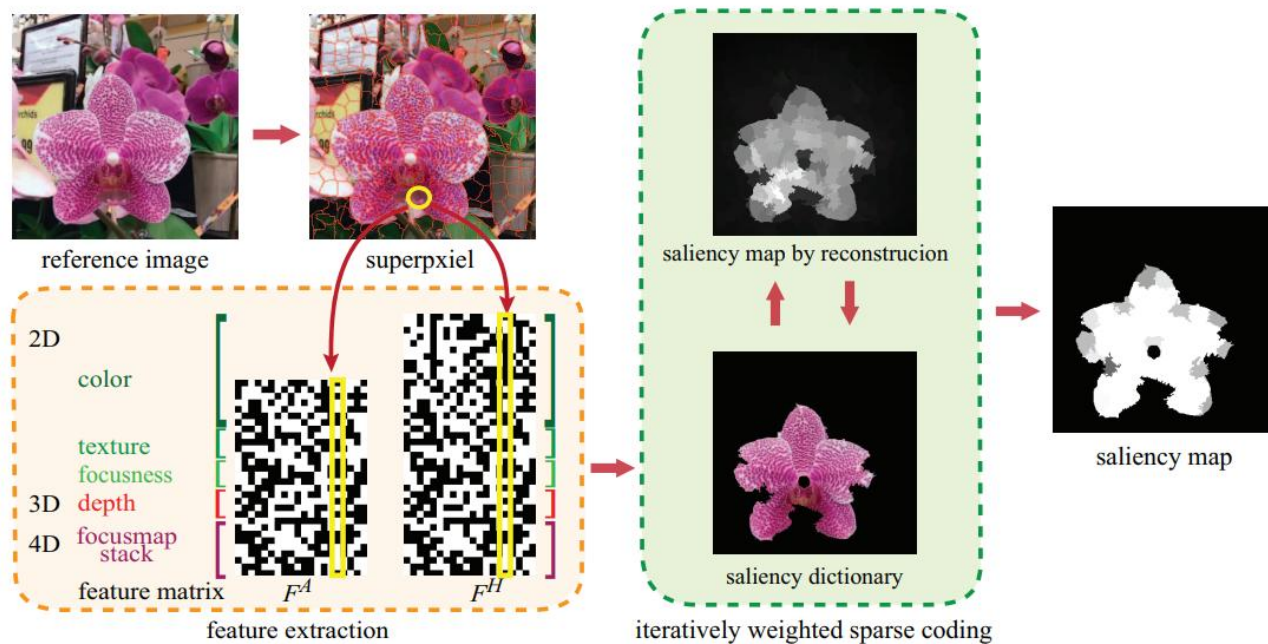
DILF method (2015)

- Combines **color, depth and focusness cues**, and **background prior** from **SLIC's superpixels**.



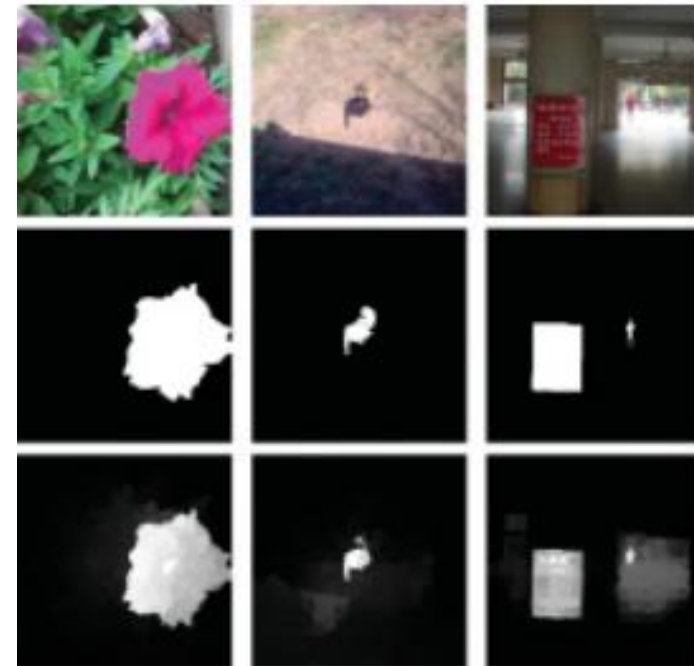
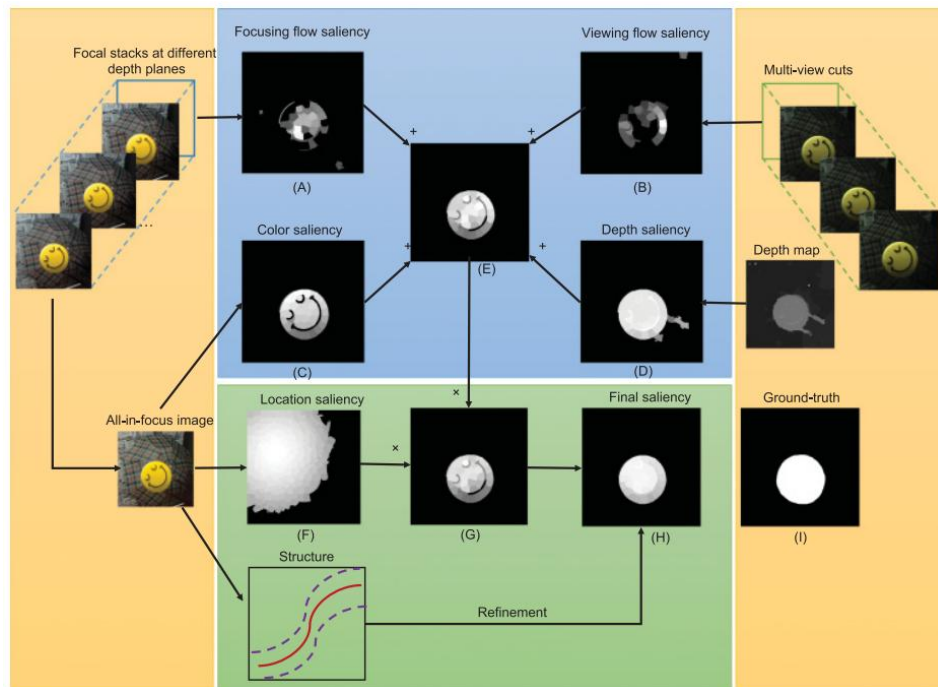
WSC method (2015)

- **Segment image into superpixels using SLIC** and extract features (color, texture, focusness, depth).
- **Reconstruct each region** using background and foreground dictionaries; high reconstruction error → likely salient.



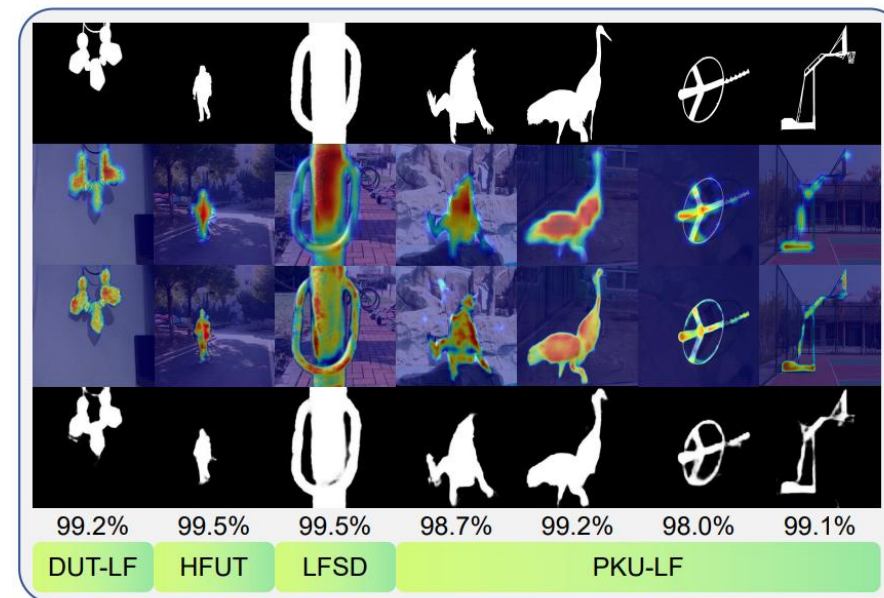
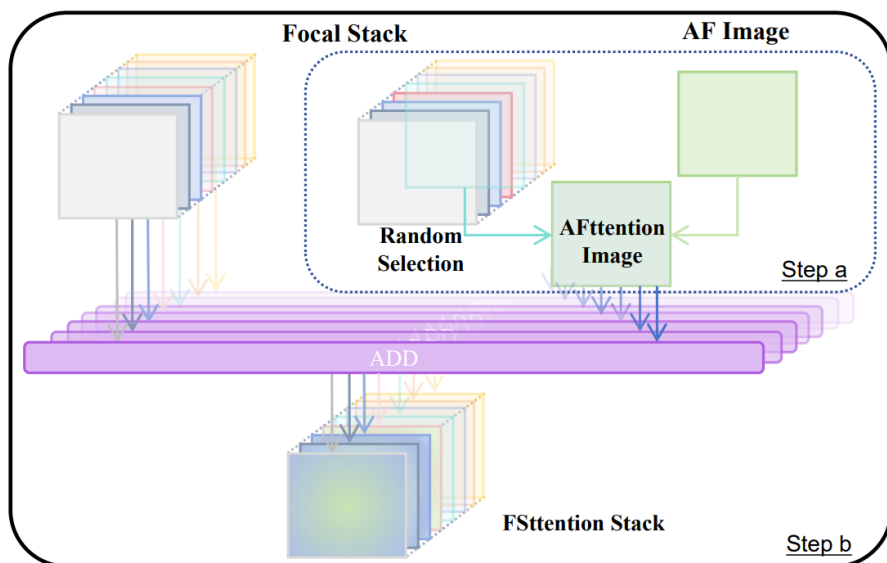
Multi-Cue method (2017)

- From SLIC's superpixels, it computes **color, depth, focusing flow, and viewing flow**.
- Measure **dissimilarity** between **superpixels** and apply **spatial priors** (center bias, background suppression).



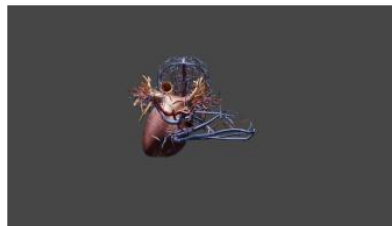
LF-Tracy method (2024)

- **Single-stage transformer** neural network to encode all-focus and focal stack images.
- Uses **MixLD** augmentation **to fuse spatial and depth cues**.
- Attention-guided feature fusion and multi-scale refinement loss.



Testing dataset: Tampere dataset

- Nine full-parallax LFs synthetically generated in Blender, **more complex than other available datasets**.
- **Complete focal stack** representing the scene at multiple focus depths.



(a) Vessels



(b) Dragon



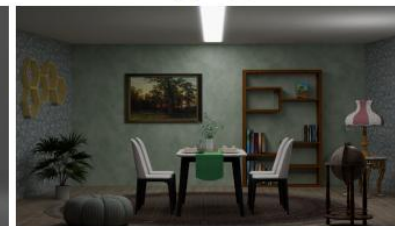
(c) Zoo



(d) Toys



(e) Laboratory



(f) Dining room



(g) Flower



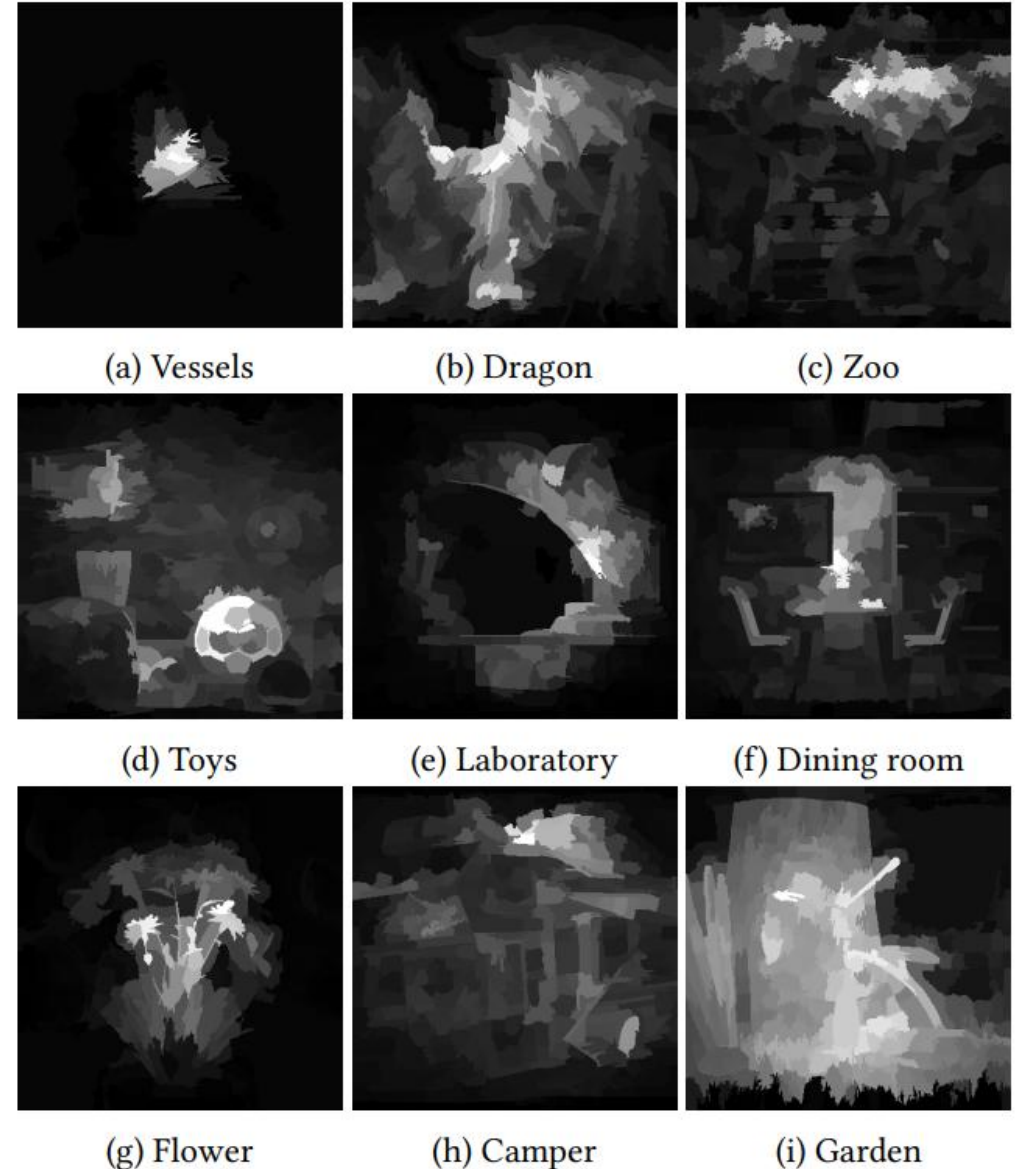
(h) Camper



(i) Garden

DILF results

- **DILF requires** images of **360×360** (all-focus images) and **1080×1080** (focal stack): **downsampled and cropped** to obtain square images.
- **Structured maps** but lost **objects border information** during cropping.



WSC results

- Performed **poorly**, failing in complex scenes with fine details or similar foreground-background
- WSC method **fails** to produce **consistent** and **reliable** saliency maps.



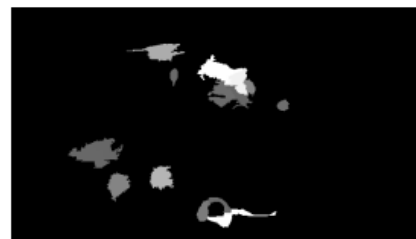
(a) Vessels



(b) Dragon



(c) Zoo



(d) Toys



(e) Laboratory



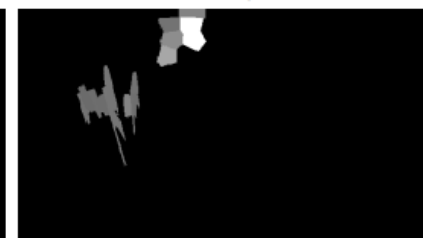
(f) Dining room



(g) Flower



(h) Camper



(i) Garden

Multi-Cue results

- Overall improvement, but **saliency detection remains inconsistent and unreliable.**
- Some **non-salient areas** are incorrectly highlighted.



(a) Vessels



(b) Dragon



(c) Zoo



(d) Toys



(e) Laboratory



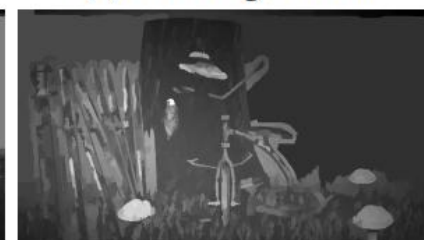
(f) Dining room



(g) Flower



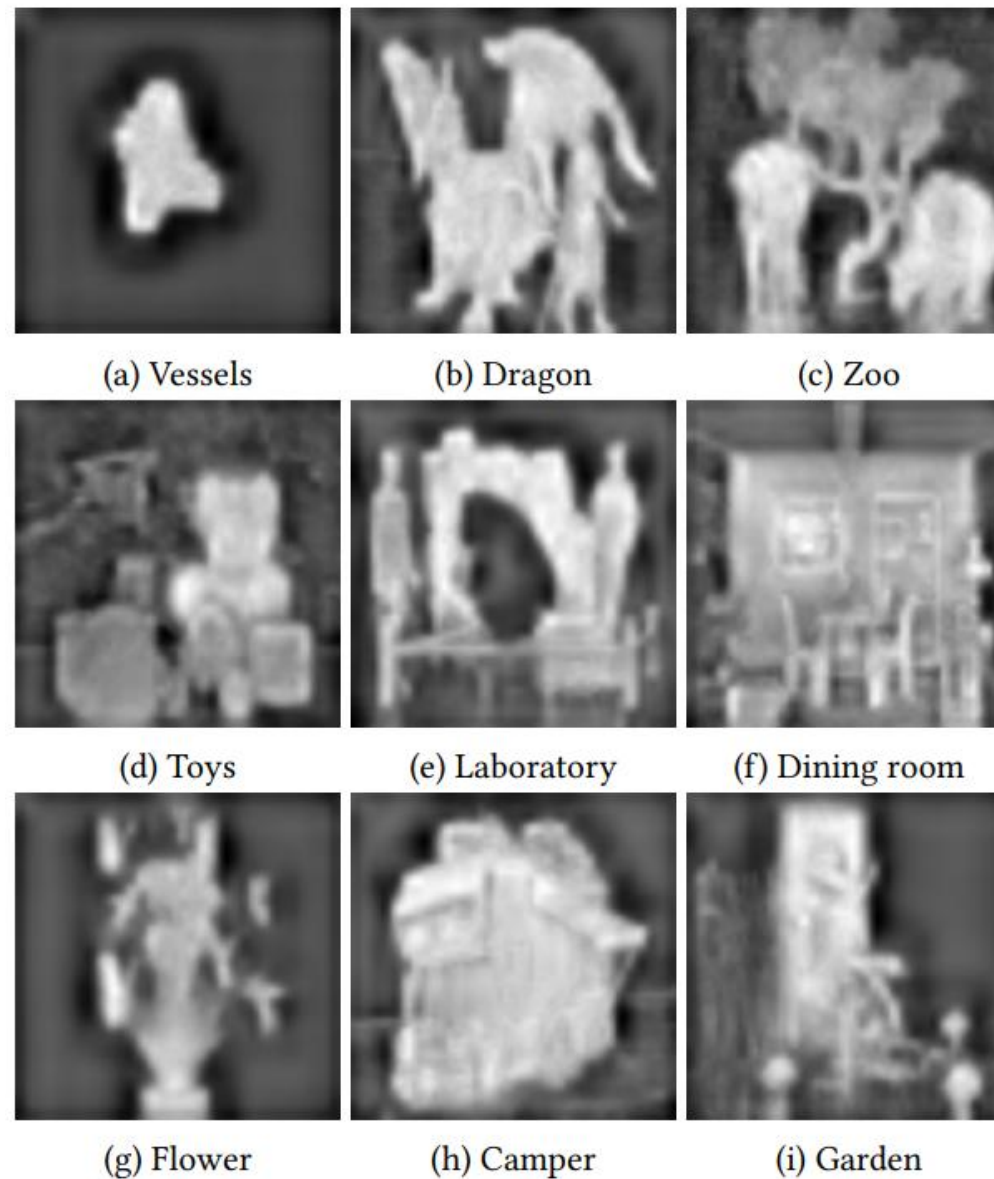
(h) Camper



(i) Garden

LF-Tracy results

- LF-Tracy's training has been reduced for hardware constraints (500 epochs -> 100).
- Showing **best generalization capabilities**, but **complexity is notable**.
- “*Object completeness bias*”: same intra-object saliency



Conclusions

- Current methods show a gap between algorithmic saliency and human perception.
- **LF-Tracy** generalizes on Tampere dataset but shows *object completeness bias* (uniform saliency across objects).
- Training deep models is limited by scarce annotated LF data and high computational cost.
- **Future work:** collect **eye-tracking data** to align saliency with human perception.



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Thank you for the attention!

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